Modeling and Validation of a Soccer Ball's Three-Dimensional Flight to the Goal

Problem B

Abstract

Gravity bends a football's path downward, drag reduces its speed, and Magnus force redirects a spinning flight. We compare gravity, gravity with drag, a constant resultant force, and gravity with aerodynamic forces. The constant-resultant model determines acceleration from three early positions. The aerodynamic model (M4) uses a constant drag coefficient and a linear spin–lift relation to estimate initial velocity, spin direction and aerodynamic parameters.

Three actual penalties supply 39 sampled frames. Three early positions per flight enter inversion, with subsequent measurements used for comparison. For M4, goal-plane errors are 0.360, 0.146 and 0.113 m for the two placement shots and the power shot; the latter has a subsequent root-mean-square path error (RMSE) of 0.093 m. A separate competition-position example quantifies parameter effects: a 1° direction change moves the crossing by about 0.20 m, and the largest grid-search shift under a 2 mm early-position constraint is 0.1052 m. Among 100,000 joint perturbations with fixed mass and radius, 95.772% clear the goal. A same-state curved-shot calculation quantifies rotational damping, while relative airflow describes wind effects.

Keywords: Soccer ball, aerodynamic drag, Magnus effect, inverse prediction, trajectory comparison

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1 Introduction

1.1 Problem Background

A penalty kick is a direct contest between a designated kicker and the opposing goalkeeper. The kicker strikes the ball from the penalty mark towards the goal. In a typical professional penalty, the ball then flies freely through the air towards the goal, while the goalkeeper must choose where and when to move. An effective shot depends on speed and placement, as well as the kicker's ability to conceal its intended direction.

The importance of penalties to match outcomes has motivated research into shooting and saving, particularly the ball's motion after it leaves the foot. Topics include curved trajectories and aerodynamic forces[3], surface structure[5, 4], and trajectory measurements and aerodynamic-parameter estimation[9]. These studies connect kicking technique with the physics of ball flight and help explain the motion of a penalty shot.

1.2 Problem Restatement

Problem B concerns the prediction of a penalty trajectory from its early flight positions. The goal is 11 m from the penalty mark, 7.32 m wide and 2.44 m high. The task is to develop and test a general method that uses the ball-centre coordinates at 0.05 s and 0.10 s after the kick to predict its most likely entry point and estimate the uncertainty of that prediction.

The preceding discussion suggests three broad stages of the ball's motion: **contact with the kicker**, **free airborne flight** and **motion after the shot**. We examine the free flight after ball–foot contact has ended and before the ball enters the goal; the effect of contact is represented by the initial velocity and spin. The forces and governing equations are discussed further in Section 3. Initial-state inference from early positions, subsequent trajectory prediction and assessment of those predictions are discussed further in Section 4. Section 5 uses the specified example to examine parameter sensitivity and the distribution of goal-plane crossings under stated perturbations, connecting the predicted location with its uncertainty. The technical workflow is summarized in Figure 1.

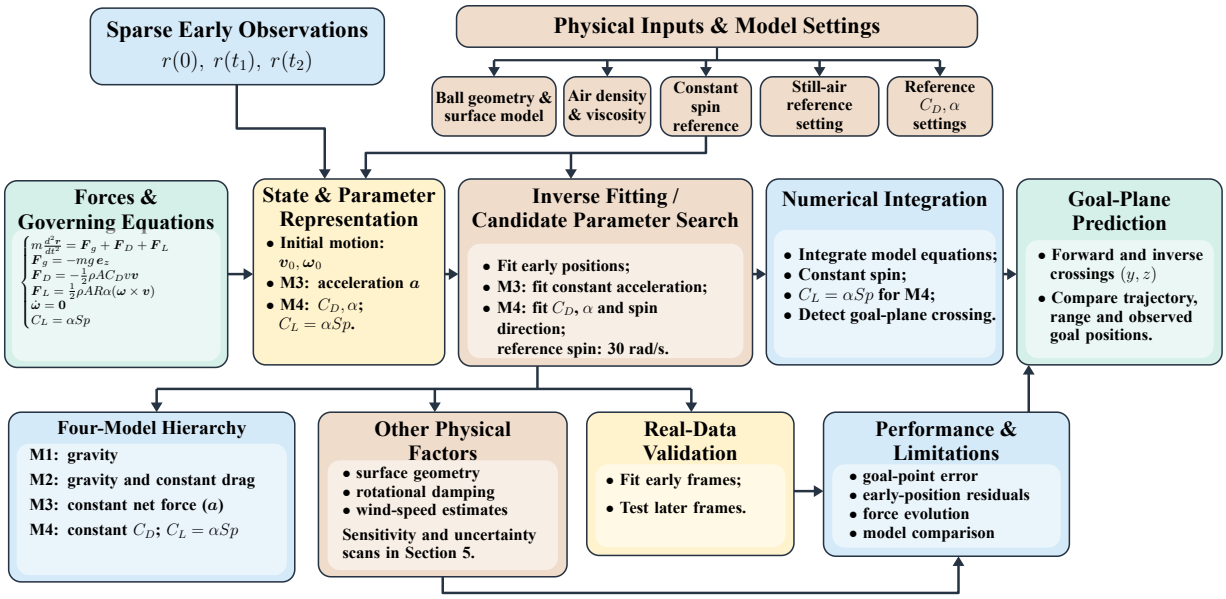


Figure 1: Three-force modeling, inverse prediction and real-penalty validation.

2 Preparation of the Models

2.1 Assumptions

1. **Free flight.** At the start of the motion considered, the ball has left the kicker's foot; contact is represented by the initial velocity and spin. We assume a proficient kick, with the ball remaining airborne without contacting the ground or another object before reaching the goal plane.

2. **Rigid spherical ball.** The ball remains spherical and its mass is uniformly distributed. Seams and asymmetric surface geometry are discussed further in Section 6.1.

3. **Uniform gravity and air properties.** The magnitude and direction of gravitational acceleration remain constant over the flight, and air density and viscosity remain uniform.

4. **Still air.** The surrounding air is treated as stationary. The effects of wind on relative airflow and aerodynamic force are discussed further in Section 6.3.

5. **Constant spin.** Aerodynamic torque is neglected, so the spin vector remains constant during flight. Rotational damping is discussed further in Section 6.2.

2.2 Notations

The symbols used below and their physical meanings are summarized in Table 1.

Table 1: Principal quantities and competition reference values.

Symbol	Meaning	Value or unit
$\boldsymbol{r}, \boldsymbol{v}$	Ball-centre position and velocity	m; m/s
$\boldsymbol{\omega}, \omega$	Spin vector and its magnitude	rad/s
m, R, A	Mass, radius and frontal area	0.43 kg; 0.11 m; πR^2
g	Gravitational acceleration	9.81 m/s^2
ρ	Air density	1.2 kg/m^3
$\nu, \mu = \rho\nu$	Kinematic and dynamic viscosity	1.50×10^{-5} m^2/s; Pa s
Re, Sp	Reynolds number and spin parameter	dimensionless
C_D, C_L, α	Drag, lift and linear-spin parameters	dimensionless
$\lambda_D, \lambda_\alpha$	Coefficient ratios to reference values	dimensionless
$\theta, \boldsymbol{e}_L$	Spin–airflow angle and Magnus direction	rad; unit vector
q, v	Dynamic pressure and ball speed	Pa; m/s
$\boldsymbol{Q}, I_b$	Damping torque and moment of inertia	N m; kg m^2

We follow Maponi and Piombin [3] for the ball parameters and Asai et al. [5] for the reference air properties, with dynamic viscosity $\mu = \rho\nu$.

3 Model Establishment

3.1 Geometric Relations

As shown in Figure 2, the local origin is the penalty mark, with x towards the goal, y horizontal and lateral, and z vertically upward. The goal plane is $x = 11$ m and the opening is $-3.66 \le y \le 3.66$ m, $0 \le z \le 2.44$ m.

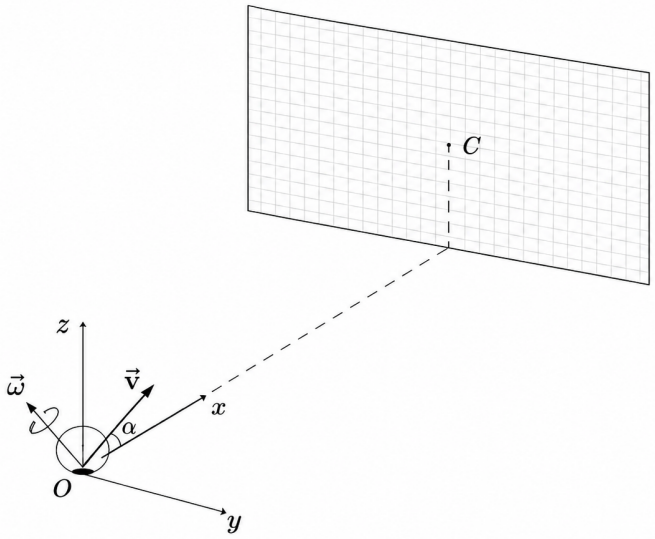


Figure 2: Penalty-kick geometry in local coordinates; O is the nominal penalty mark and C an illustrative goal-plane point.

The entire ball clears the opening when its centre satisfies $|y| \leq 3.66 - R$ and $R \leq z \leq 2.44 - R$.

3.2 Force Analysis

As shown in Figure 3, we consider gravity, buoyancy, drag and Magnus force in turn.

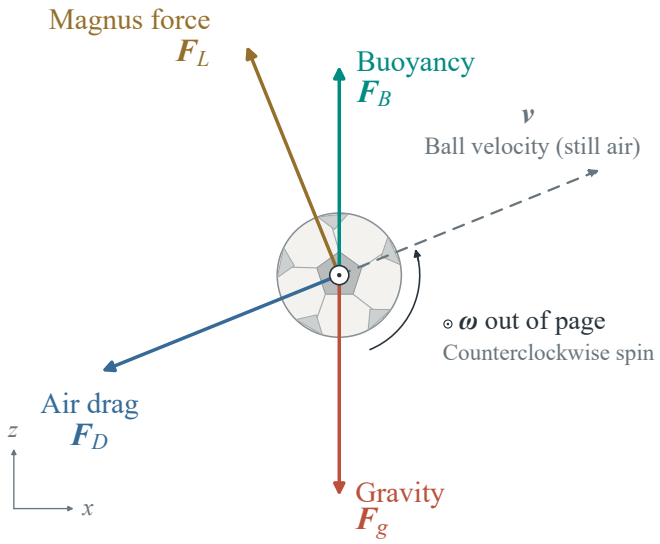


Figure 3: Schematic of gravity, buoyancy, drag and Magnus force on the football.

3.2.1 Gravity

Let $\boldsymbol{e}_z$ be the upward vertical unit vector. For mass m in the uniform field of gravitational acceleration magnitude g,

$$\boldsymbol{F}_g = -mg\boldsymbol{e}_z, \tag{1}$$

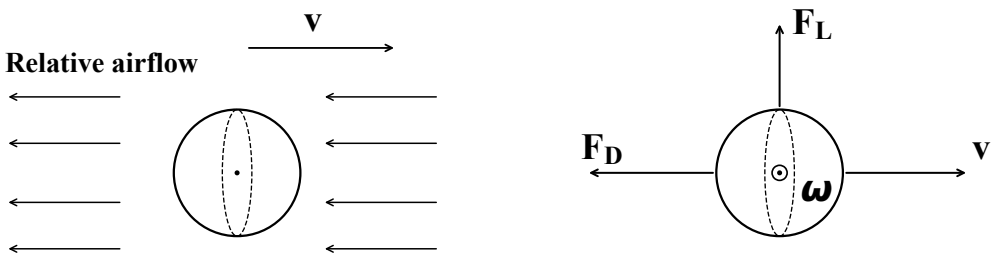
By Archimedes' principle, the displaced air also exerts an upward buoyancy force $\boldsymbol{F}_B$. For ball volume $V_b = \frac{4}{3}\pi R^3$ and air density ρ,

$$\boldsymbol{F}_B = \rho V_b g\, \boldsymbol{e}_z = \frac{4}{3}\pi R^3 \rho g\, \boldsymbol{e}_z. \tag{2}$$

From these expressions, the combined vertical force is $\boldsymbol{F}_g + \boldsymbol{F}_B = -(m - \rho V_b)g\boldsymbol{e}_z$. Buoyancy is 1.56% of weight for the reference parameters and is therefore omitted as a small correction in the subsequent calculations.

3.2.2 Aerodynamic Drag

Let $\boldsymbol{v}$ be the ball-centre velocity. Since the air is assumed stationary, the incident airflow in the ball frame has velocity $-\boldsymbol{v}$. As shown in Figure 4, the airflow slows near the forward face of the ball.



(a) Airflow in the ball frame **(b) Example: spin out of the page**

Figure 4: Schematic of relative airflow and force directions.

Let p_∞ be the incident static pressure and p_s the pressure at the forward stagnation point. Bernoulli's relation along an idealized incident streamline is

$$p_\infty + \frac{1}{2}\rho v^2 = p_s.$$

The pressure rise defines the dynamic pressure q:

$$q = p_s - p_\infty = \frac{1}{2}\rho v^2.$$

Following Goff and Carré [9], aerodynamic drag depends on air density, relative speed, frontal area and the ball's flow and surface characteristics. The latter enter through the drag coefficient C_D, so the drag magnitude is

$$F_D = C_D q A.$$

Since drag opposes the relative motion, substitution of the dynamic pressure leads to

$$\boldsymbol{F}_D = -\frac{1}{2}\rho A C_D v \boldsymbol{v}. \tag{3}$$

Here $A = \pi R^2$ is the frontal area and ρ is air density. We approximate C_D as constant during the ball's flight.

3.2.3 Magnus Lift

A spinning football changes the surrounding airflow and experiences a transverse Magnus force. As shown in Figure 5, rotation makes the relative flow asymmetric. Let $\boldsymbol{\omega}$ be the angular-velocity vector.

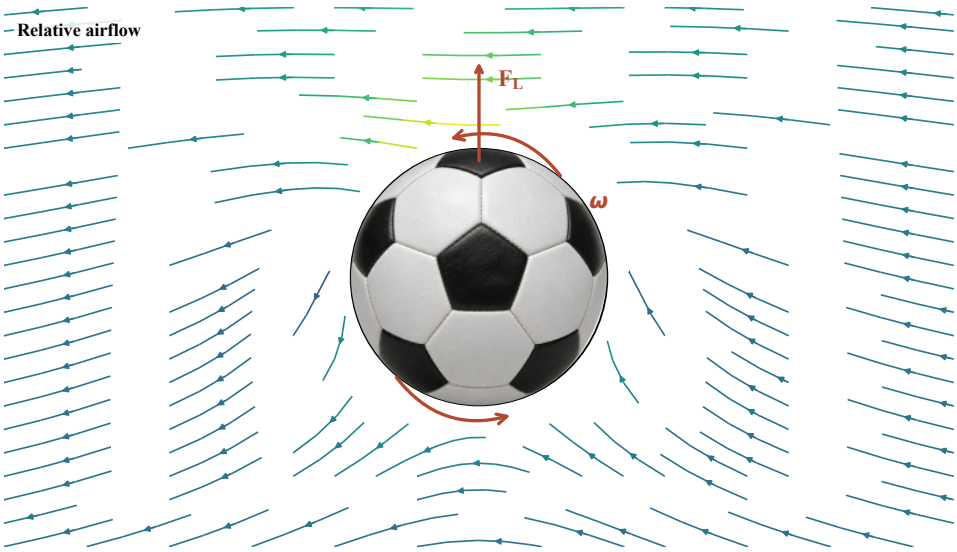


Figure 5: Idealized flow around a spinning football.

Following Goff and Carré [9], transverse lift is expressed in terms of dynamic pressure and frontal area. When the spin axis makes angle θ with the relative velocity, the lift magnitude is

$$F_L = qAC_L \sin\theta.$$

Here C_L is the transverse lift coefficient and θ is the spin–airflow angle. We approximate the relation between lift coefficient and spin parameter as linear:

$$C_L = \alpha \frac{R\omega}{v}.$$

Here α is dimensionless. Based on Kiratidis and Leinweber [6], we approximate this coefficient as constant during the flight. Substituting the dynamic-pressure and linear-spin relations,

$$\boldsymbol{F}_L = \frac{1}{2}\rho A R \alpha (\boldsymbol{\omega} \times \boldsymbol{v}). \tag{4}$$

3.3 Dynamical Equations

Since the football is treated as a rigid body, its motion can be separated into translation of the centre of mass and rotation about that centre. Let $\boldsymbol{L}$ be angular momentum about the centre, I_b the moment of inertia, and $\boldsymbol{Q}$ the resultant external torque. Centre-of-mass translation follows Newton's second law,

$$m\dot{\boldsymbol{v}} = \sum \boldsymbol{F} \tag{5}$$

while rotation about the centre follows the angular-momentum theorem,

$$\frac{d\boldsymbol{L}}{dt} = \boldsymbol{Q}, \qquad \boldsymbol{L} = I_b\boldsymbol{\omega}. \tag{6}$$

From zero applied torque and constant spherical inertia, $I_b\dot{\boldsymbol{\omega}} = \boldsymbol{0}$, so the spin remains constant. Combining this result with the three retained forces,

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ \dot{\boldsymbol{\omega}} = \boldsymbol{0}. \end{cases} \tag{7}$$

Let $\boldsymbol{r}_0$, $\boldsymbol{v}_0$ and $\boldsymbol{\omega}_0$ be the initial centre position, velocity and angular velocity, respectively; their subscripts x, y, z denote components in the chosen axes. The initial conditions are

$$\begin{cases} \boldsymbol{r}(0) = \boldsymbol{r}_0, \\ \boldsymbol{v}(0) = (v_{x0}, v_{y0}, v_{z0}), \\ \boldsymbol{\omega}(0) = (\omega_{x0}, \omega_{y0}, \omega_{z0}). \end{cases} \tag{8}$$

4 Model Solving

This chapter develops models and an algorithm for inferring the initial motion from early ball positions and predicting the goal-plane crossing.

4.1 Solution Method

4.1.1 Model selection

Based on the force analysis in Chapter 3, we compare four models with different force assumptions:

1. **M1: gravity.** With $\boldsymbol{a} = (0, 0, -g)$, only the three components of $\boldsymbol{v}_0$ are estimated. Spin has no effect on the trajectory.

2. **M2: gravity and constant-coefficient drag.** Only weight and drag opposite to velocity are considered. The estimated parameters are the initial velocity and constant C_D, for four unknowns in total.

3. **M3: constant resultant force.** All forces are represented by a constant resultant acceleration $\boldsymbol{a}$. The three components of $\boldsymbol{v}_0$ and three of $\boldsymbol{a}$ form six unknowns. Gravity is included in this resultant.

4. **M4: gravity, drag and Magnus force.** The force relations of Section 3 determine the motion. We estimate three initial velocity components, two spin-direction angles, C_D and the linear lift parameter α: seven unknowns.

4.1.2 Inverse method

For measured positions $\boldsymbol{r}_i$ at times t_i ($i = 0, 1, 2$), let $\widehat{\boldsymbol{r}}(t_i; \boldsymbol{p})$ denote the prediction associated with the unknown parameter vector $\boldsymbol{p}$. We minimize the six coordinate residuals through

$$J(\boldsymbol{p}) = \sum_{i=1}^{2} \left\| \widehat{\boldsymbol{r}}(t_i; \boldsymbol{p}) - \boldsymbol{r}_i \right\|^2 .$$

The parameter bounds for the four models appear in Appendix A.

After fitting, the inferred state is integrated to the goal plane $x = 11$ m. The goal-crossing distance error is

$$E_g = \sqrt{(\widehat{y}_g - y_g)^2 + (\widehat{z}_g - z_g)^2}, \tag{9}$$

where (y_g, z_g) is the observed crossing and hats denote the prediction. Further parameter and source records appear in Appendix A.

4.1.3 Dataset

To assess the general applicability of the models, we use the Sportec Open DFL dataset described by Bassek et al. [1] and released by DFL [2], which combines match events with 25 Hz multi-camera tracking. We select the following three penalties, referred to as Cases 1, 2 and 3:

1. **Case 1:** Dejan Ljubicic's penalty in 1. FC Köln–FC Bayern München on 27 May 2023.
2. **Case 2:** Philipp Klement's penalty in Fortuna Düsseldorf–1. FC Kaiserslautern on 11 November 2022.
3. **Case 3:** Dawid Kownacki's penalty in Fortuna Düsseldorf–SSV Jahn Regensburg on 26 August 2022.

The selected starting positions are $\boldsymbol{r}_0 = (3.04, 0.55, 0.13)$ m, $(2.57, 0.85, 0.34)$ m and $(0.77, 0.08, 0.24)$ m for Case 1, Case 2 and Case 3, respectively; each defines $t = 0$. As shown in Table 2, the next two positions occur at 0.04 and 0.08 s.

Table 2: Two later input positions and observed goal crossings; coordinates in m.

Case	t (s)	(x, y, z)	Observed (y_g, z_g)
Case 1	0.04	$(4.19, 0.82, 0.15)$	$(2.7500, 0.1600)$
Case 1	0.08	$(5.25, 1.07, 0.18)$	
Case 2	0.04	$(3.64, 1.20, 0.41)$	$(3.0697, 0.4657)$
Case 2	0.08	$(4.71, 1.53, 0.47)$	
Case 3	0.04	$(1.99, 0.34, 0.46)$	$(2.9188, 1.1789)$
Case 3	0.08	$(3.18, 0.62, 0.65)$	

4.2 Case Analysis

According to Hunter et al. [11], penalty technique and ball speed affect accuracy and the goalkeeper's anticipation. Studying the kicker's technique helps interpret and predict penalty trajectories. We distinguish two common approaches:

1. **Precise placement shot.** A side-foot contact favours control of direction and accurate placement. A low trajectory can carry the ball towards a lateral target while keeping it close to the turf.

2. **Power instep shot.** An instep contact emphasizes a fast strike, shortening the goalkeeper's available response time. The launch direction can produce a higher flight towards the goal.

Reviewing the match footage, we classify Case 1 and Case 2 as placement shots and Case 3 as a power shot. As shown in Figure 6, Figures 6a and 6b illustrate the low placement and higher power-shot flight patterns, respectively.



(a) Illustrative placement shot.



(b) Illustrative power shot.

Figure 6: Photographic examples of placement and power shots.

4.2.1 Precise placement shots

As shown in Figure 7, both placement shots remain low.

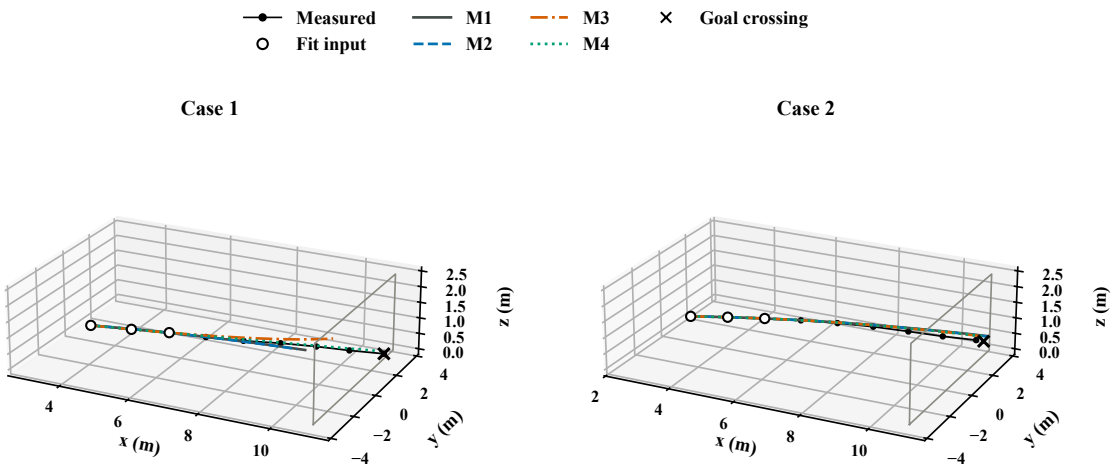


Figure 7: Measured and predicted trajectories for the two placement shots.

In M4, the initial vertical velocities are 0.55 and 1.89 m/s for the two cases. As shown in Figures 7 and 8, M1 and M2 touch the ground early in Case 1, and M3 does not reach the goal plane, while M4 stays closer to the measured flight. Comparing the four models suggests that the upward Magnus force associated with spin partly offsets gravitational fall and allows the ball to continue towards the goal along a low trajectory. In Case 2, both M3 and M4 reproduce bending in the y direction, with M4 providing the most accurate prediction. Thus, the small upward velocity keeps a placement shot low, while Magnus force plays a key role in maintaining its height.

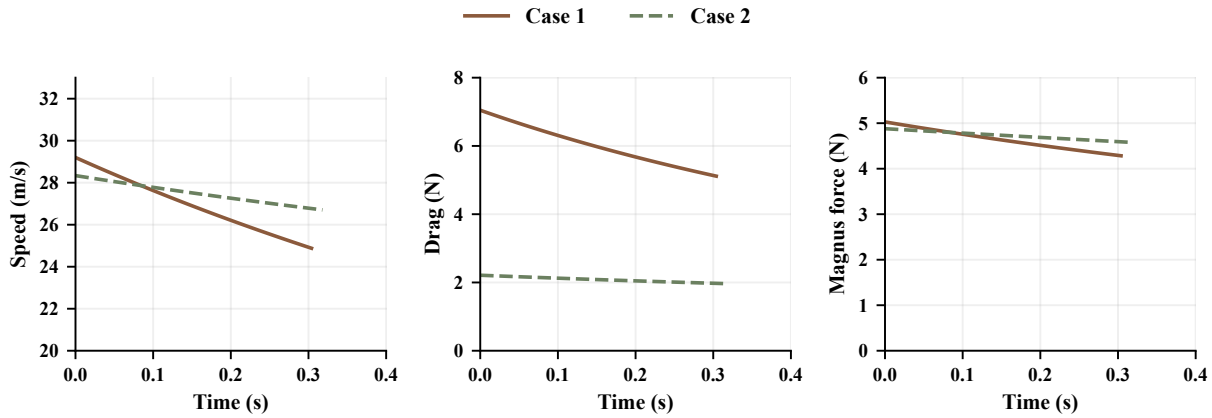


Figure 8: M4 speed, drag and Magnus-force magnitude for the two placement shots. Axis scales match Figure 10.

4.2.2 Power instep shot

As shown in Figure 9, Case 3 rises substantially higher than the two placement shots.

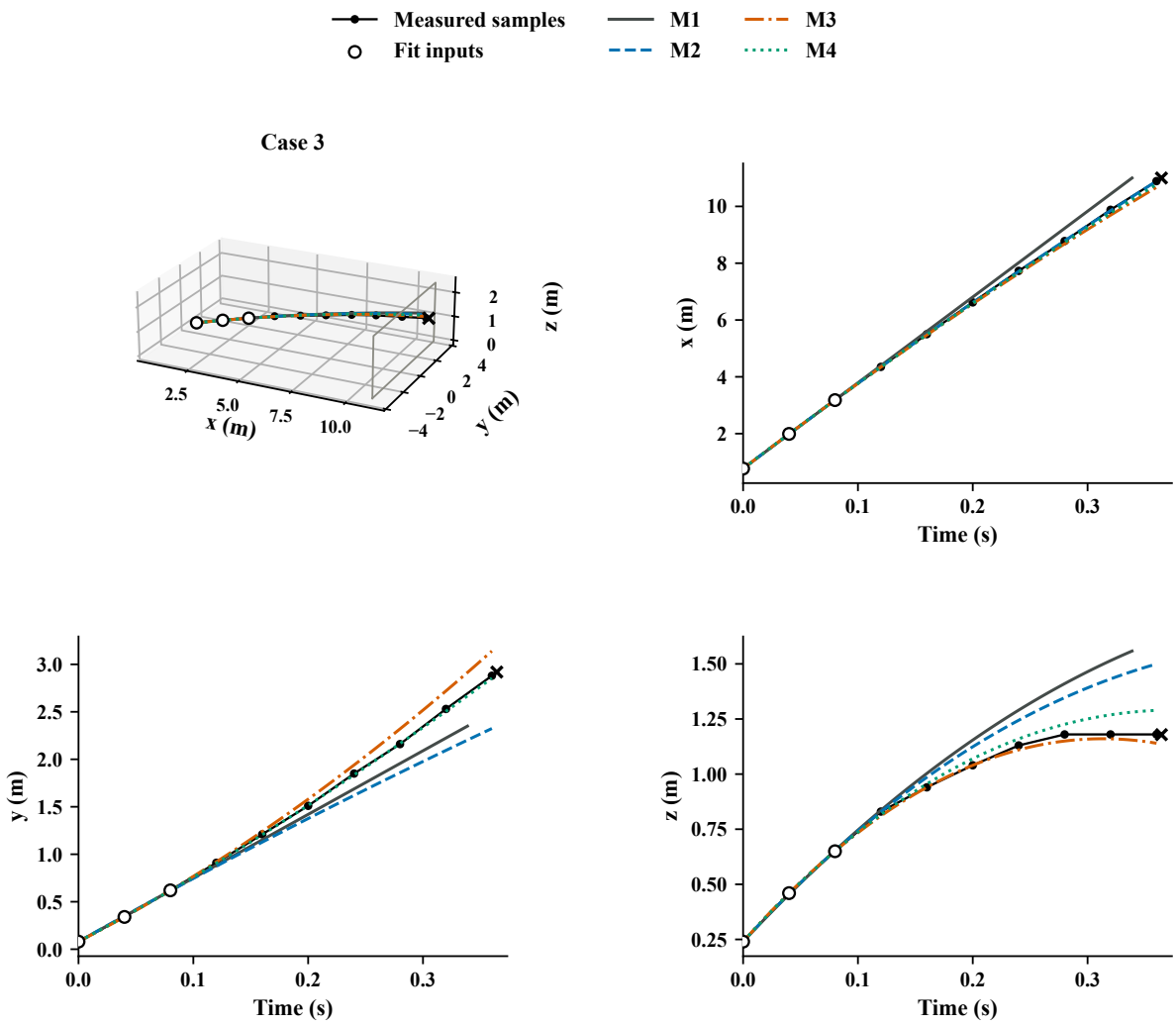


Figure 9: Case 3: spatial trajectory, $x(t)$, $y(t)$ and $z(t)$ in a 2×2 layout. Symbols follow Figure 7.

The M4 initial speed is 32.05 m/s, exceeding the two placement values of 29.20 and 28.34 m/s. Its elevation is 10.46° and initial vertical velocity 5.82 m/s, producing a higher flight. As shown in Figure 10 and Table 3, M1 and M2 do not adequately capture the lateral bending. The M3 and M4 trajectories show how the upward initial velocity raises the ball, while gravity and the downward Magnus component in M4 bend the flight back. Magnus force therefore plays a key role in the lateral curvature of this power shot.

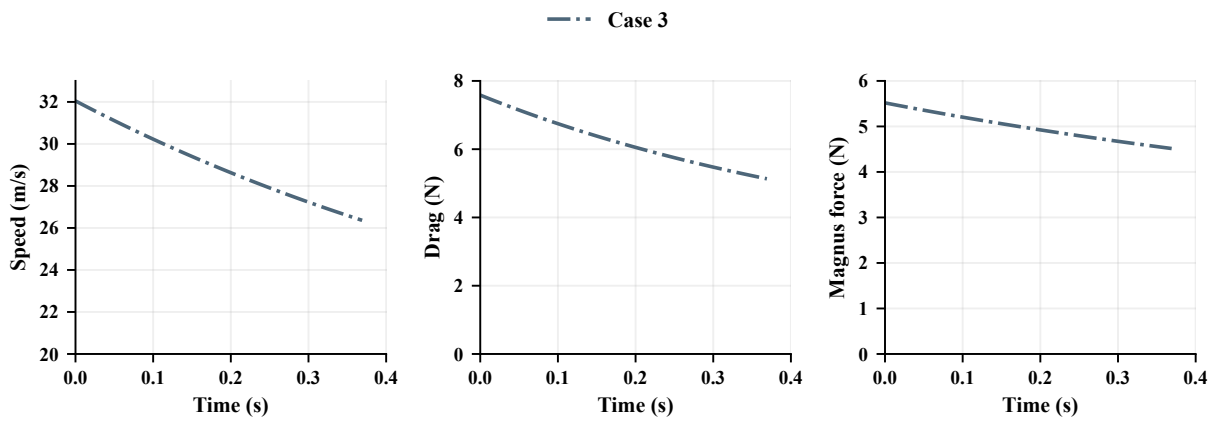


Figure 10: M4 speed, drag and Magnus-force magnitude for the power shot, on the same axes as Figure 8.

Table 3: Power-shot goal crossings and distance errors (m).

Case	Model	Observed (y, z)	Predicted (y, z)	Distance
Case 3	M1	$(2.919, 1.179)$	$(2.350, 1.558)$	0.684
Case 3	M2	$(2.919, 1.179)$	$(2.349, 1.506)$	0.657
Case 3	M3	$(2.919, 1.179)$	$(3.289, 1.126)$	0.374
Case 3	M4	$(2.919, 1.179)$	$(2.934, 1.291)$	0.113

Considering both a valid goal-plane prediction and its distance error, accounting for drag and spin, especially the changing Magnus force during flight, makes **M4 the better model for these three cases.**

4.3 M4 Inversion for the Problem B Observations

The initial centre is $(0, 0, 0.11)$ m. As shown in Table 4, Problem B specifies two later positions at 0.05 and 0.10 s. We apply the same M4 inverse method and parameter settings to these inputs.

Table 4: Initial position and two observations specified in Problem B.

t (s)	x (m)	y (m)	z (m)
0	0	0	0.11
0.05	1.50	0.45	0.44
0.10	2.96	0.86	0.74

The inferred initial motion is

$$\begin{aligned} \boldsymbol{v}_0 &= (30.3784, 9.4074, 6.8974)\ \mathrm{m/s}, \\ \boldsymbol{\omega}_0 &= (19.7434, 1.2060, -22.5553)\ \mathrm{rad/s}. \end{aligned} \tag{10}$$

As shown in Figure 11, **the predicted goal-plane crossing is** $(y_g, z_g) = (2.6159, 1.9526)$ **m**. The inferred speed of 32.54 m/s, elevation of 12.24° and pronounced lateral bending indicate that **this trajectory more closely resembles the power-shot type studied in this chapter.**

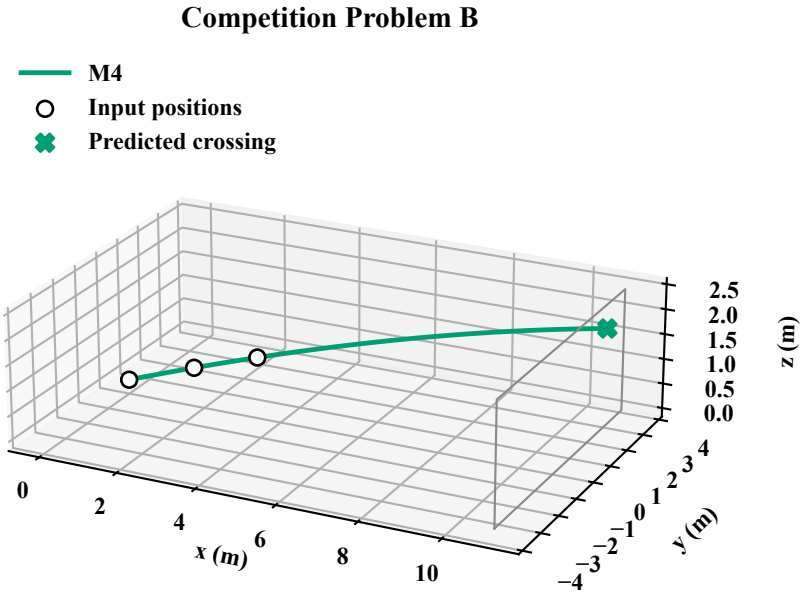


Figure 11: M4 prediction for Problem B, showing the three input positions, goal and predicted crossing.

Section 5 uses this inferred state to study changes in the goal crossing and its distribution.

5 Sensitivity Analysis

The M4 state inferred in Section 4.3 is the reference for this chapter. As shown in Table 4, its early positions remain the basis for the position-constrained analysis. Mass and radius remain fixed at $m = 0.43$ kg and $R = 0.11$ m. Write the reference crossing as $(y_g^*, z_g^*) = (2.6159, 1.9526)$ m and $t_g^* = 0.398967$ s.

5.1 Effects of individual parameters

One parameter changes at a time. Speed varies by $\pm 5\%$; spin magnitude, C_D, α and air density each vary by $\pm 10\%$; velocity and spin-axis azimuth and elevation each vary by $\pm 2°$. Each scan uses 41 equally spaced values, giving 369 flights. Define

$$\begin{aligned} \Delta y_g &= y_g - y_g^*, \quad \Delta z_g = z_g - z_g^*, \quad \Delta t_g = t_g - t_g^*, \\ D_g &= \sqrt{(\Delta y_g)^2 + (\Delta z_g)^2}. \end{aligned} \tag{11}$$

As shown in Table 5, we compare direction changes of $\pm 1^\circ$ and other changes of $\pm 5\%$, taking the larger D_g and $|\Delta t_g|$ from each positive–negative pair.

Table 5: Goal-plane displacement and arrival-time change under one-factor perturbations.

Parameter	Perturbation	max D_g (cm)	max $\lvert\Delta t_g\rvert$ (ms)
Launch speed	$\pm 5\%$	8.356	20.659
Velocity azimuth	$\pm 1^\circ$	20.070	1.965
Velocity elevation	$\pm 1^\circ$	20.218	1.692
Spin magnitude	$\pm 5\%$	3.873	0.287
Spin-axis azimuth	$\pm 1^\circ$	1.002	0.071
Spin-axis elevation	$\pm 1^\circ$	0.852	0.084
Drag coefficient	$\pm 5\%$	0.620	2.274
Lift parameter	$\pm 5\%$	3.873	0.287
Air density	$\pm 5\%$	4.142	1.990

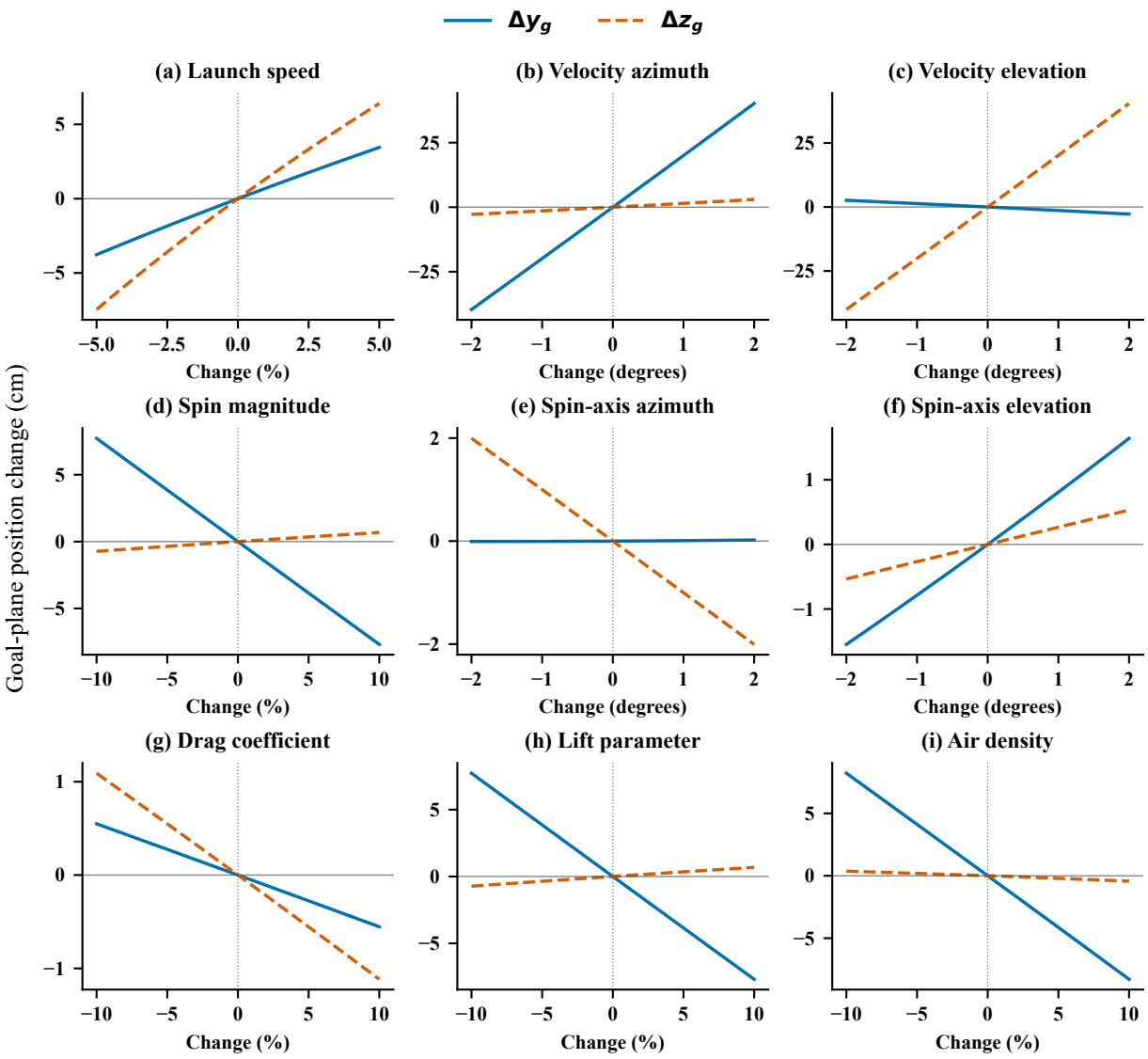


Figure 12: Goal-plane coordinate responses to nine one-factor perturbations. Blue solid curves show Δy_g and orange dashed curves Δz_g, in cm. Vertical scales differ between panels.

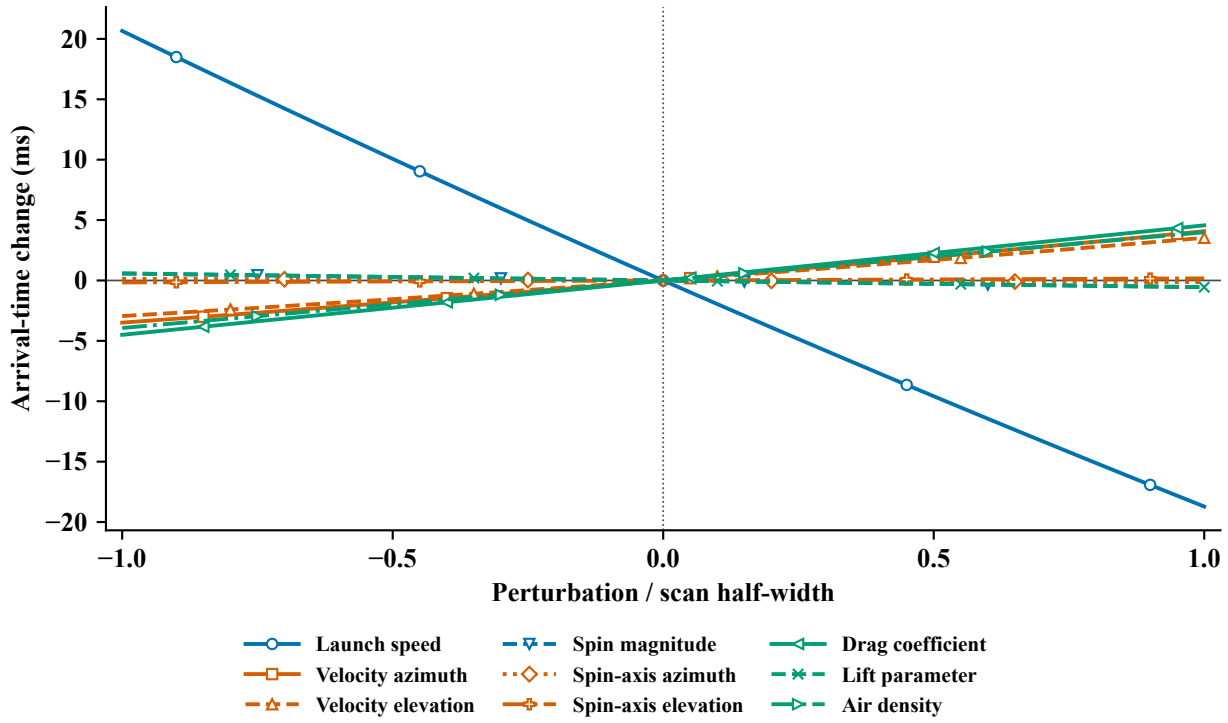


Figure 13: One-factor arrival-time responses. Perturbation divided by scan half-width is plotted horizontally, so ± 1 marks each scan endpoint. The vertical axis is Δt_g in ms; positive values mean later arrival.

As shown in Figure 12, a 1° change in velocity azimuth or elevation produces a maximum displacement of 20.070 cm or 20.218 cm. As shown in Figure 13, a 5% speed change produces 8.356 cm and 20.659 ms. Speed sets the time available to reach the goal, while direction directly changes lateral and vertical displacement; drag further alters flight time through deceleration.

The linear lift law contains the product of α and spin magnitude. Applying the same relative change to either quantity therefore produces coincident response curves. For any positive dimensionless scale factor k,

$$\boldsymbol{\omega}_0' = k\boldsymbol{\omega}_0, \qquad \alpha' = \frac{\alpha}{k} \quad \Longrightarrow \quad \boldsymbol{F}_L' = \boldsymbol{F}_L. \tag{12}$$

This compensation explains the reference spin convention and connects the individual scans with the position-constrained analysis below.

5.2 Crossing variation under early-position constraints

As shown in Table 4, the early positions remain fixed while velocity and spin are refitted to examine aerodynamic parameter combinations. With the coefficient references C_D^*, α^*, define $\lambda_D = \dfrac{C_D}{C_D^*}$ and $\lambda_\alpha = \dfrac{\alpha}{\alpha^*}$. Each multiplier takes 17 values in $[1.2, 1.8]$. At each node, three velocity and three spin components are fitted, using total speed at most 35 m/s, spin components at most 60 rad/s in magnitude, and maximum early-coordinate residual at most 2 mm.

The grid produces 153 compatible flights; 63 lie in the inner square $[1.35, 1.65]^2$. As shown in Table 6, the compatible flights span a range of crossings and arrival times. The largest crossing displacement in the wider scan is 0.1052 m.

Table 6: Ranges of goal-plane crossings compatible with the early positions.

Range of each multiplier	y_g (m)	z_g (m)	t_g (s)
$[1.35, 1.65]$	2.5538–2.6550	1.9122–2.0307	0.3973–0.4025
$[1.2, 1.8]$	2.5478–2.6601	1.9075–2.0368	0.3973–0.4043

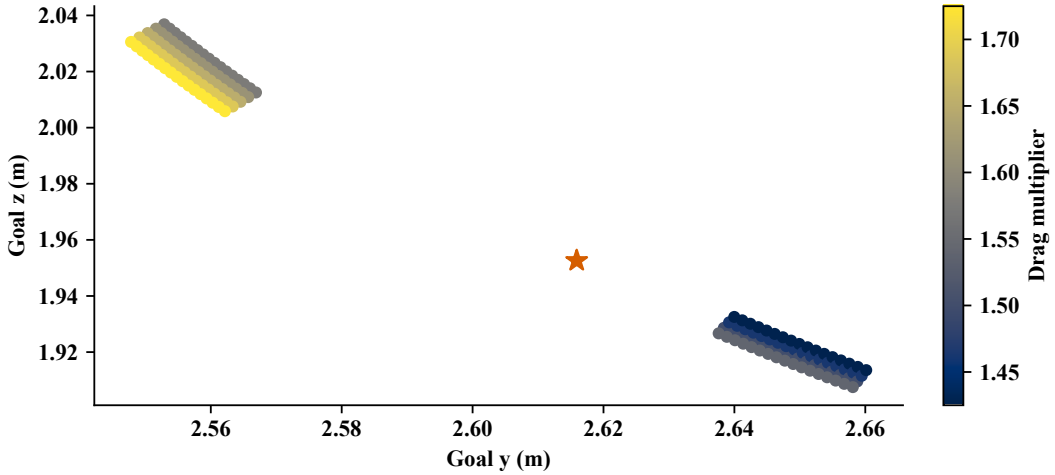


Figure 14: Grid-search crossings under the 2 mm early-position constraint. Colour indicates λ_D; the orange star is the reference flight.

As shown in Figure 14, different forces and initial states can stay close at early times and separate later, producing the lateral and vertical spread shown here. Refitting the other inverse quantities connects this spread to compensation under the position constraints.

5.3 Joint perturbations and Monte Carlo simulation

Nine parameters are sampled independently and uniformly over the full ranges in Section 5.1. Seed 20261108 generates 100000 trajectories. Mass, radius and initial position remain fixed while launch motion and aerodynamic coefficients vary together. Trajectories use fourth-order Runge–Kutta with a 0.002 s step and cubic interpolation of the crossing event. Every trajectory reaches the goal plane before ground contact. As shown in Table 7, the simulated positions and arrival times have both sampled ranges and central percentile intervals.

Table 7: Monte Carlo crossing positions and arrival times (100,000 trajectories).

Output	Sample minimum–maximum	Central 95% interval, one coordinate
y_g (m)	1.9633–3.2143	2.1858–3.0505
z_g (m)	1.4381–2.4646	1.5534–2.3535
t_g (s)	0.3685–0.4346	0.3783–0.4218

The central 95% intervals are the 2.5th–97.5th percentiles of each output. The mean crossing is $(2.6170, 1.9511)$ m, with lateral and vertical standard deviations 0.2455 m and 0.2360 m. The largest displacement from the reference crossing is 0.801 m.

For a joint region, let $\boldsymbol{q}_i = (y_{g,i}, z_{g,i})^T$, with sample mean $\bar{\boldsymbol{q}}$ and covariance $\widehat{\Sigma}$. Let d_i^2 be the dimensionless squared distance from the mean and $c_{0.95}$ its empirical 95th percentile. The joint region $\mathcal{E}_{0.95}$ is defined by

$$\begin{aligned} d_i^2 &= (\boldsymbol{q}_i - \bar{\boldsymbol{q}})^T \widehat{\Sigma}^{-1} (\boldsymbol{q}_i - \bar{\boldsymbol{q}}), \\ \mathcal{E}_{0.95} &= \{\boldsymbol{q} : (\boldsymbol{q} - \bar{\boldsymbol{q}})^T \widehat{\Sigma}^{-1} (\boldsymbol{q} - \bar{\boldsymbol{q}}) \le c_{0.95}\}. \end{aligned} \tag{13}$$

From the empirical cutoff $c_{0.95} = 4.5916$, the ellipse contains 95% of the simulated crossings.

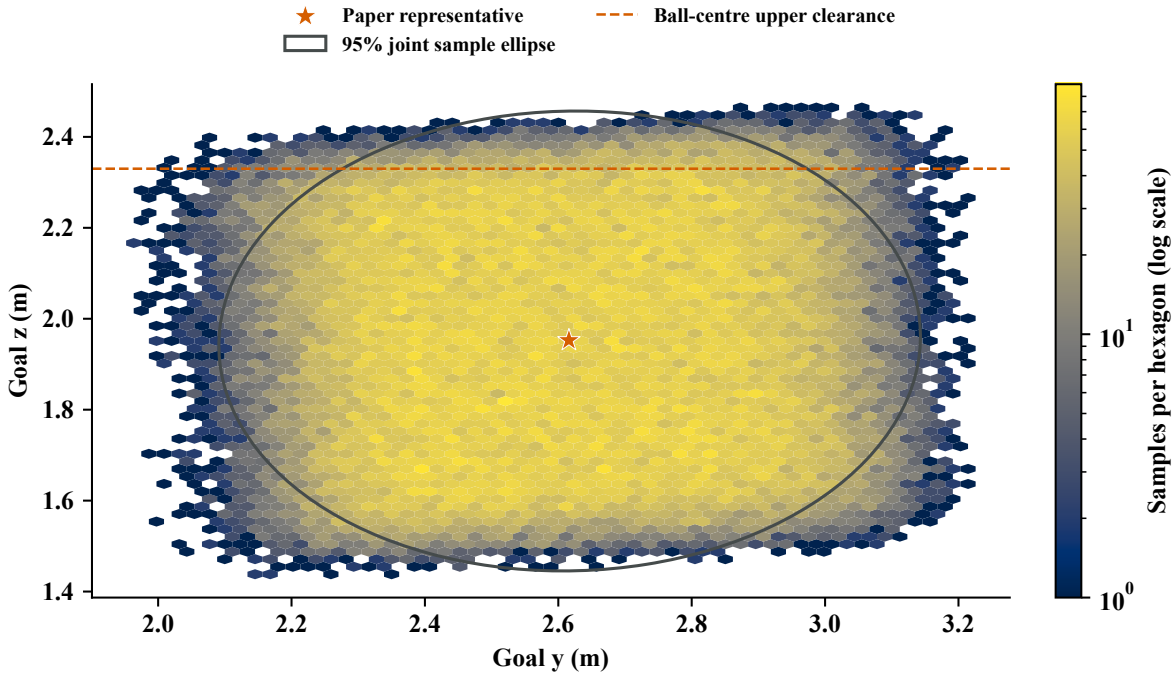


Figure 15: Goal-plane crossings under 100,000 joint perturbations. Hexagon colour indicates sample count, the ellipse indicates the joint 95% region, and the dashed line marks the centre-height clearance limit $z_g = 2.33$ m.

As shown in Figure 15, the joint spread places some crossings above the opening. From the clearance rule of Section 3, 95772 flights (95.772%) pass fully through the opening and 4228 (4.228%) cross too high.

As shown in Figure 16, 400 randomly selected flights connect the input changes with their outcomes. Each line connects nine inputs to y_g, z_g, t_g. Inputs use their perturbation limits and outputs the full-sample extrema for linear scaling to 0–1. Blue lines clear the opening, orange lines are too high, and the grey dashed line is the reference state.

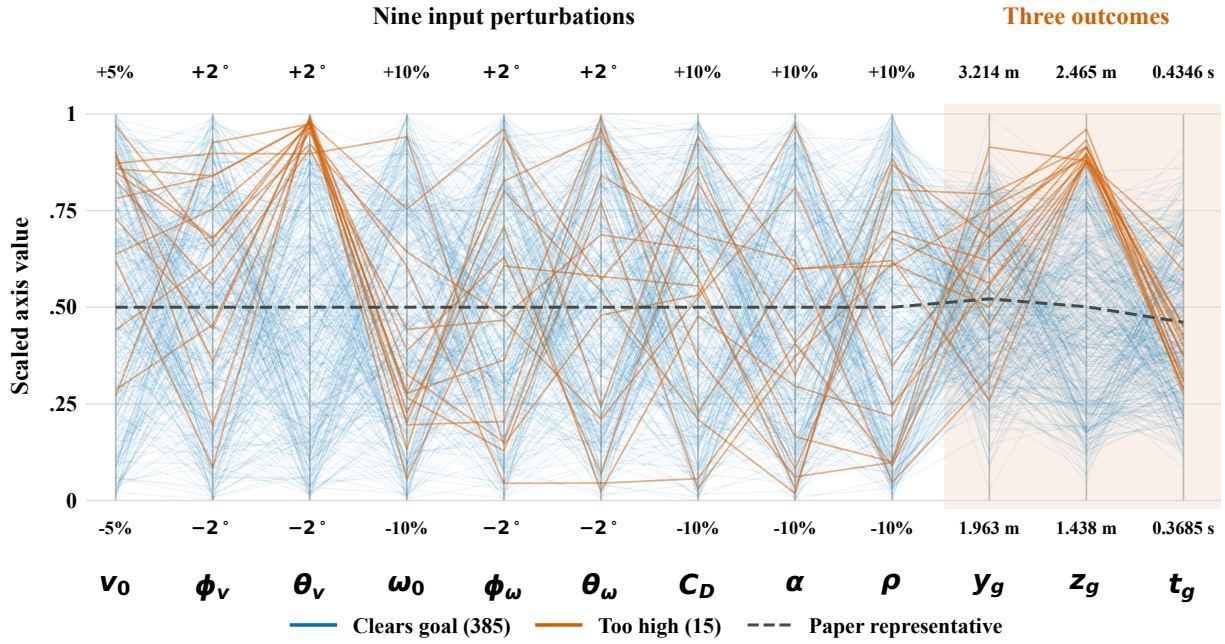


Figure 16: Parallel coordinates linking inputs, goal-plane positions and arrival time. The first nine axes are inputs; three shaded axes are outputs. The displayed subset contains 385 clear flights and 15 high flights.

Following a line reveals the elevation, spin, coefficients and arrival time associated with a high crossing. The spread on the output axes shows lateral and vertical changes together.

As shown in Figure 17, increasing the sample from 50000 to 100000 changes the central-95% interval endpoints by at most 0.264 mm laterally, 0.506 mm vertically and 0.138 ms in time. Two independent seeds each produce 10000 flights. Their largest interval-endpoint differences from the full sample are 5.425 mm laterally, 4.320 mm vertically and 0.328 ms in time.

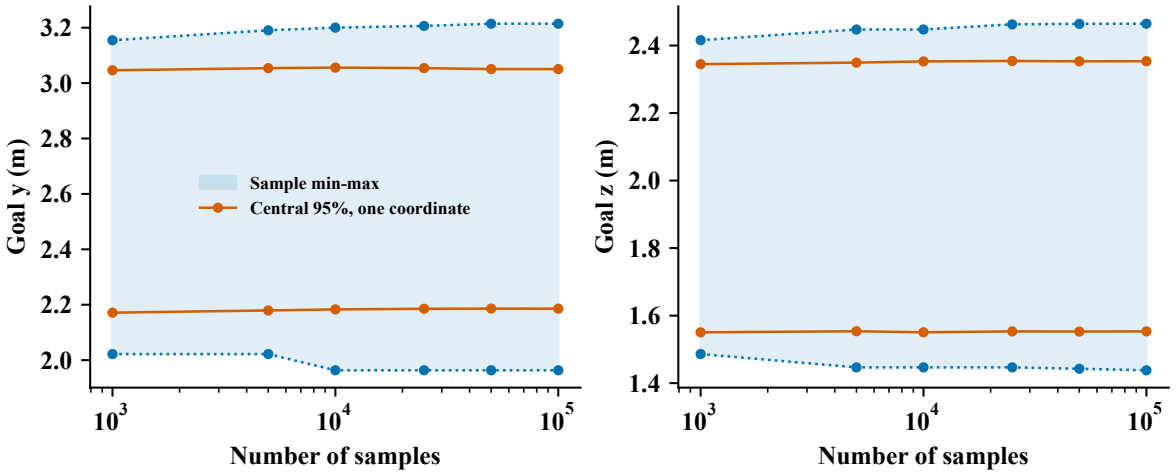


Figure 17: Sample-count effects on crossing ranges and percentile intervals. Blue shading and dotted curves show sampled extrema; orange curves show each coordinate's central 95% interval.

6 Discussion of Other Factors

The main model isolates gravity, drag and Magnus lift. We now examine the physical effects omitted from its geometry, spin evolution and ambient airflow.

6.1 Asymmetric Surface Geometry

According to Paudel and Yan [4], full-flight simulations distinguish realistic football geometry from a smooth sphere. As shown in Figure 18, the comparison includes both idealized geometry and actual panel textures; as shown in Figure 18a, we distinguish the seamed and smooth geometries. For launch speeds of 10, 15 and 20 m/s and spin rates of 31.5, 63 and 94.5 rad/s, their results show that seams and surface structure alter separation, wake dynamics and Magnus force. The smooth sphere produces greater lateral bending in their nine comparisons, and vertical flight-time differences become more visible at higher speed and rotation.



(a) Illustrative seamed-football and smooth-sphere comparison, following Paudel and Yan [4].



Fig. 1. Vintage and Adidas Fevernova balls



Fig. 2. Adidas Teamgeist 2006 and 2009 balls



Fig. 3. Adidas 2010 Jabulani and Nike 2010 T-90 balls

(b) Six footballs and surface details, reproduced from Figures 1–3 of Alam et al. [10].

Figure 18: Football geometry, panels and surface textures.

According to Alam et al. [10], wind-tunnel measurements reveal different drag coefficients for six footballs and a smooth sphere over a range of Reynolds numbers. As shown in Figure 18b, these footballs have different panels and textures; as shown in Figure 19, their drag coefficients also differ. The differences illustrate the influence of ball surface structure on aerodynamic drag.

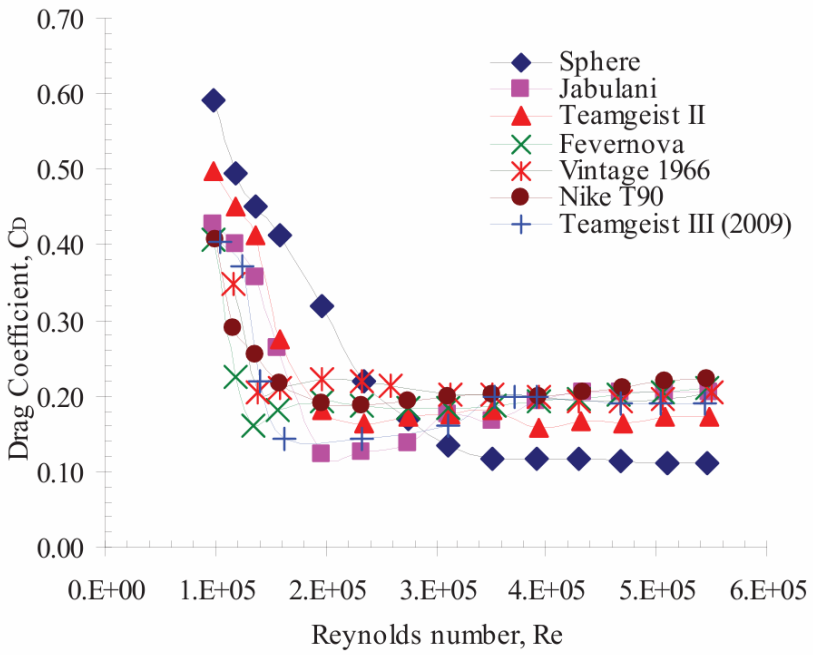


Figure 19: Drag coefficient against Reynolds number for different footballs and a smooth sphere, reproduced from Figure 5 of Alam et al. [10].

The spherical geometry in the present report describes the centre-of-mass motion and projected area. Its force coefficients come from football measurements, so the average surface influence enters through C_D and C_L. Explicit panel orientation and asymmetric wake fluctuations provide a further level of description when lateral-force details are important. These published results support retaining football-specific coefficients in a spherical trajectory model.

6.2 Rotation and Aerodynamic Damping

Aerodynamic torque opposes spin, gradually weakening rotation and changing the Magnus force. Following Grad et al. and the MuJoCo model [7, 8], we use high-Reynolds-number rotational damping: stronger spin produces stronger resistance. The dimensionless angular-drag parameter $C_{\text{angular}} = 1.5$ is the software reference value. Following Maponi and Piombin [3], the inertia is approximated by that of a thin spherical shell. Viscous damping is much weaker than the inertial contribution in this example, so the comparison focuses on the latter.

The reference curved shot has initial velocity $(23.6354, 0, 4.1676)$ m/s, spin $(0, 0, 35)$ rad/s and unit force corrections. Holding this state fixed, we compare constant spin with rotational damping. As shown in Figure 20, decreasing spin slightly changes the flight curvature and goal-plane crossing.

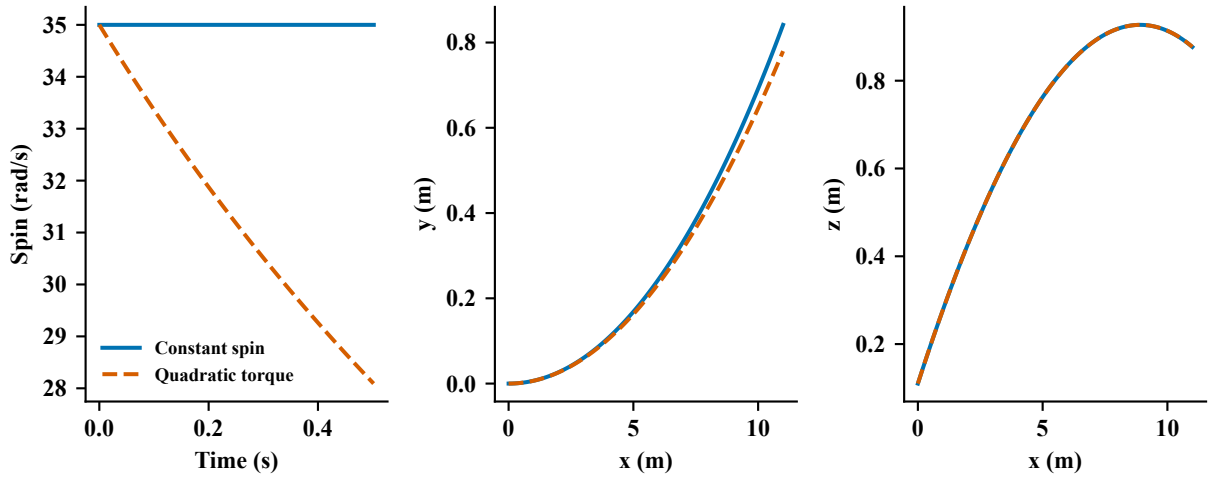


Figure 20: Constant spin and rotational damping under the same initial curved-shot conditions.

Spin falls by 19.740%, while the crossing moves 0.0616 m (0.560% of the 11 m distance) and arrival time changes by -0.066%. Rotational damping therefore has a small endpoint effect in this short reference flight.

6.3 Wind Speed and Force Estimates

Following the relative-airflow treatment of Maponi and Piombin [3], we introduce air-relative velocity $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$ in this subsection, where $\boldsymbol{v}_{\text{wind}}$ is air velocity and $u = \|\boldsymbol{u}\|$ is relative speed. For wind, $\boldsymbol{u}$ and u replace $\boldsymbol{v}$ and v in the aerodynamic relations of Section 3. Let w be wind speed in m/s and v the ball speed. At fixed coefficients, denote the still-air dynamic pressure by q_0 and the headwind, tailwind and crosswind values by q_{head}, q_{tail} and q_{cross}. From the squared relative-speed dependence,

$$\frac{q_{\text{head}}}{q_0} = (1 + \frac{w}{v})^2, \qquad \frac{q_{\text{tail}}}{q_0} = (1 - \frac{w}{v})^2, \qquad \frac{q_{\text{cross}}}{q_0} = 1 + (\frac{w}{v})^2. \tag{14}$$

Crosswind rotates the airflow by $\arctan(\frac{w}{v})$ and changes the force direction. As shown in Table 8, illustrative winds of 1, 3 and 5 m/s alter dynamic pressure at the 24 m/s curved-shot speed. Let $F_{L,0}$, $F_{L,\text{head}}$ and $F_{L,\text{tail}}$ denote lift magnitudes in still air, headwind and tailwind. From Eq. (4), the M4 linear lift law depends on the first power of relative speed. With fixed spin and $0 \le w < v$,

$$\frac{F_{L,\text{head}}}{F_{L,0}} = 1 + \frac{w}{v}, \qquad \frac{F_{L,\text{tail}}}{F_{L,0}} = 1 - \frac{w}{v}. \tag{15}$$

Table 8: Illustrative wind at the 24 m/s curved-shot launch: dynamic-pressure changes at fixed coefficients.

w (m/s)	Headwind (%)	Tailwind (%)	Crosswind (%)	Angle (°)
1	+8.51	-8.16	+0.17	2.39
3	+26.56	-23.44	+1.56	7.13
5	+46.01	-37.33	+4.34	11.77

A 5% limit on the headwind dynamic-pressure change corresponds to $\frac{w}{v} \leq \sqrt{1.05} - 1$, or $w \leq 0.59$ m/s at this launch. The still-air approximation is most suitable for indoor or weak-wind flight; stronger wind should enter through the relative-velocity calculation.

7 Conclusions

The four models describe gravitational fall, deceleration, a constant resultant force and rotational deflection. M3 obtains constant acceleration from three positions. M4 inserts constant C_D and $C_L = \alpha S p$ into the motion equations, updating drag and Magnus force as speed changes.

For M4, the crossing errors are 0.360, 0.146 and 0.113 m for the three penalties. Its power-shot crossing error and subsequent path RMSE are the smallest among the four models. For Case 1, M3 passes through the early positions but reaches a maximum $x = 10.973$ m, short of the goal; M4 supplies a goal-plane prediction for this low flight.

In the competition-position example, a 1° launch-direction change moves the crossing by about 0.20 m, while speed strongly affects arrival time. Spin magnitude and α compensate through the linear lift law. The largest grid-search shift under the 2 mm position constraint is 0.1052 m; 95.772% of 100,000 jointly perturbed flights clear the opening.

Rotational damping shifts the reference curved-shot crossing by 0.0616 m. Wind enters through relative velocity: along the flight direction, drag changes with the squared speed ratio and linear Magnus force with the speed ratio.

8 Model Evaluation

8.1 Model Strengths

Physical interpretation. The model connects the path to three familiar effects: gravity bends it downward, drag reduces speed, and Magnus force redirects a spinning ball. The four-model comparison separates these mechanisms from a constant-acceleration description, making the differences between low placement shots and a higher power shot easier to explain.

A common framework for inference and prediction. Three early positions provide the input for estimating initial motion and continuing the flight with the same governing equations. Speed, force histories and goal-plane crossings can therefore be interpreted together. The common coordinate system and termination rules also make comparisons between models and recorded shots consistent.

Comparison with measurements. Published aerodynamic measurements support the coefficient choices, while later positions from three actual penalties assess the predicted continuation.

The error table, trajectory plots and sensitivity study link the physical assumptions to observable differences. The wind and rotational-damping comparisons further identify when the simpler flight assumptions remain useful.

8.2 Model Weaknesses

Sensitivity of the inferred state. The fitted velocity, spin direction and aerodynamic parameters depend on the early tracking positions, frame selection and search bounds. Position differences over short time intervals strongly influence estimated acceleration, especially for flights close to the ground. The reference spin magnitude and the coupling between spin and lift parameter also affect the selected M4 state.

Simplified aerodynamics and environment. Constant coefficients during a flight smooth over flow transitions, panel orientation and fluctuating wakes. The spherical, still-air and constant-spin assumptions likewise average effects that may become more important for stronger winds, sustained spin or longer flight. Their relative importance changes with the ball and operating conditions.

Limited coverage of recorded conditions. The three penalties cover a modest range of shot types, balls and environments. Broader comparison would benefit from ball-specific aerodynamic data, directly measured spin and more early tracking positions. These additions would strengthen the initial-state estimate and help determine when wind and rotational damping should enter the main prediction.

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Appendix A: Data Sources, Frame Selection and Parameter Bounds

Following the dataset description by Bassek et al. [1] and the DFL release [2], we retain the source event and frame identifiers. The three selected penalties contain 39 records, with 15, 12 and 12 for the respective events. As shown in Table 9, each case is associated with its player and selected samples.

Table 9: Correspondence between case labels and source records

Case	Player	Selected samples
Case 1	Dejan Ljubicic	3, 4, 5
Case 2	Philipp Klement	3, 4, 5
Case 3	Dawid Igor Kownacki	2, 3, 4

The first selected position is the time origin and the fitting times are 0, 0.04 and 0.08 s. Their source-time origins are 0.08, 0.08 and 0.04 s in the table's order. Repeated and post-goal records are retained; later positions are used only for validation. The observed goal crossing uses a source frame or interpolation. Let $(X, Y, Z_{\text{original}})$ denote the published pitch-centred coordinates. The height is interpreted in metres as ball-centre height, and the local coordinates follow from $x = X - 41.5$, $y = Y$ and $z = Z_{\text{original}}$.

As shown in Table 2, the retained inputs and observed crossings share one local coordinate system. As shown in Table 10, the inverse variables have specified search ranges.

Table 10: Component bounds and force-correction settings for inversion.

Quantity	Search interval
v_{x0} (M1, M2, M4)	$[0, 35]$ m/s
v_{y0}, v_{z0} (M1, M2, M4)	$[-35, 35]$ m/s
$\|\boldsymbol{\omega}_0\|$ (M4)	30 rad/s
Azimuth, elevation (M4)	$[-\pi, \pi], [-\frac{\pi}{2}, \frac{\pi}{2}]$ rad
$\frac{C_D}{C_D^*}, \frac{\alpha}{\alpha^*}$	$[0.5, 1.5]$

As shown in Table 11, the M4 states differ in speed, coefficients and unit spin axis $\boldsymbol{n} = \dfrac{\boldsymbol{\omega}}{\|\boldsymbol{\omega}\|}$.

Table 11: Fitted M4 states at the selected starting frames.

Event	v_0 (m/s)	C_D	α	n_x	n_y	n_z
Case 1	29.20	0.3623	2.2877	0.2280	-0.9728	0.0413
Case 2	28.34	0.1208	2.2877	0.1439	-0.2711	-0.9517
Case 3	32.05	0.3238	2.2877	-0.2276	0.3170	0.9207

Reproduction Code

The following excerpts evaluate the force components and state derivatives used in Eq. (7). Initial-state inference and subsequent trajectory prediction use these same physical relations and coefficient settings.

```
def state_derivative(t, state, config):
    """The single nine-state RHS used for prediction and inverse fitting."""
    f = forces(state[3:6], state[6:9], config)
    return np.r_[state[3:6], sum(f.values())/config.mass,
                 rotational_torque(state[6:9], config)/config.moment_of_inertia]
```

```
def forces(velocity, spin, config):
    """Return force components on ball, in newtons."""
    v = np.asarray(velocity, dtype=float) - np.asarray(config.wind, dtype=float)
    omega = np.asarray(spin, dtype=float)
    speed = np.linalg.norm(v)
    weight = (config.mass * np.asarray(config.constant_acceleration, dtype=float)
              if config.constant_acceleration is not None else
              np.array([0., 0., -config.mass * config.gravity if config.use_gravity else
                  0.]))
    drag = np.zeros(3)
    magnus = np.zeros(3)
    if speed > 1e-12:
        cd, cl, _, _, _ = coefficients(velocity, spin, config)
        q_area = 0.5 * config.air_density * np.pi * config.radius**2 * speed**2
        if config.use_drag:
            drag = -q_area * config.drag_scale * cd * v / speed
        cross = np.cross(omega, v)
        norm = np.linalg.norm(cross)
        if config.use_magnus and norm > 1e-12:
            # The literature fit is for transverse spin. Project its effect
            # onto the plane perpendicular to airflow for an oblique spin axis.
            sin_angle = norm / (np.linalg.norm(omega) * speed)
            magnus = q_area * config.lift_scale * cl * sin_angle * cross / norm
    return {"gravity": weight, "drag": drag, "magnus": magnus}
```